Lead 16.7: isogeny oracles in the algebraic group model

Round 4 of the isogeny-fibers study · 2026-10-11 · theory round with a native toy sanity check
Source note: `RESULTS.md` (full statements and proofs) · raw output: `out_seed7.json`

Status labels. Results called Theorem, Proposition, Lemma or Corollary are proved here or in `RESULTS.md`. *Heuristic* means the class-factor uniformity heuristic of §14.1. *Conjecture* means not proved. Numbers in §6 are **measured** (exact counts on one toy curve pair) or **derived** (ratios of measured counts). Toy curves only; no sub-rho claim; no claim about a deployed curve.

1. Verdict

- **Model.** AGM-Iso(Ψ) is defined in §2. The adversary reads coordinates and may compute anything. It also has oracles for a fixed set Ψ of rational isogenies and their duals across one class, and every point it *submits to an oracle* must come with a representation over the points it has received.
- **Theorem 1.** Any such adversary runs without oracles, with exactly the same success probability. The additive cost is $O(Q(\ell + \log r))$, and **no isogeny is ever evaluated**. The simulator needs only the class-joint instance (P_j, xP_j). **Theorem 2** gives the converse, so the oracles are worth exactly the transported challenges.
- **Corollary 3.** If transport along a spanning tree is computable, oracle-DLP on E_1 reduces to plain DLP on E_1. The loss is $|J|-1$ tree evaluations plus $O(Q(\ell + \log r))$. Non-tree isogenies, even of huge degree, enter only through a scalar.
- **Limits.** For DLP the AGM is vacuous without oracles: every algorithm outputs a scalar, so every algorithm is algebraic. The model has no lower-bound content and **cannot separate index calculus from rho**. Without the algebraic restriction on queries, the oracle becomes a transport oracle for factor-base points (Prop. 5). Simulating it then costs $Q \times$ (degree-dependent evaluation).
- **Summation polynomials.** Isogenies push decompositions forward exactly (Thm 6). **The kernel-saturation law of §14.1(b) is the image law of §14.1(a) on the image curve** (Thm 7). Image bases need no oracle (Prop. 8).
- **Constants vs exponents.** The oracle can only choose the representative: oracle-DLP $\equiv$ CJ-DLP $\le \min_j$ DLP(E_j). "Constants only" is **false in general**, because GHS classes over F_{2^n} with composite n contain both weak and non-weak members. It is **Conjecture 10** over prime fields.
- **Sanity check (measured).** The simulator answered 5000/5000 algebraic queries correctly after one evaluation of φ. All fiber-sum, injective and saturated identities held, with 0 mismatches. The exact equality $D_m(G) = D_m(G')$ held (6195 = 6195; 360 = 360). The heuristic ratios came out at 1.97–2.05 against a prediction of 2, and at 0.509 and 0.255 against predictions of 1/2 and 1/4.

Lead 16.7: oracle simulation in the isogeny-oracle AGM (diagram of the proof, not data)

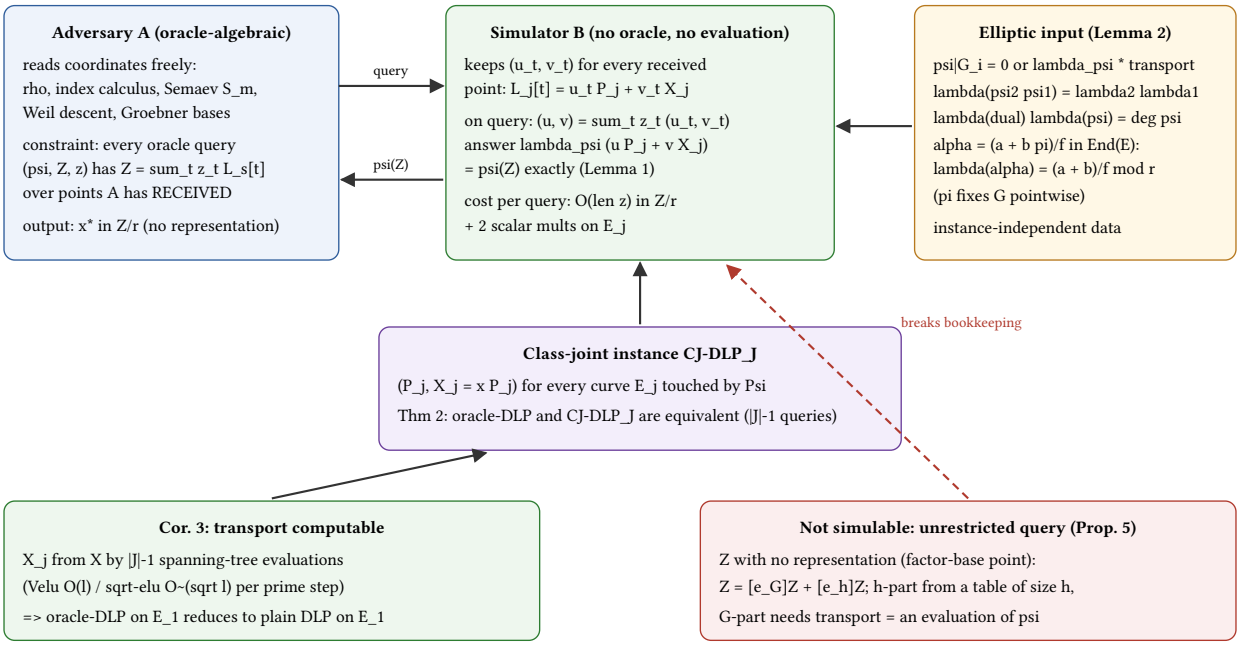


Figure 1: Diagram of the reduction of Theorem 1, the class-joint instance (Thm 2), the computable-transport branch (Cor. 3) and the unrestricted query that breaks the bookkeeping (Prop. 5). This is a diagram of the proof, not data. For this theory round it is the required visual.

2. Setting and model

F_q is a finite field. $E_1, \ldots, E_k$ are ordinary curves in one F_q-isogeny class with $N = h \cdot r$, r prime, $r \nmid h$. $G_j = E_j(F_q)[r]$ has generator P_j. $\tau_{ij}: G_i \longrightarrow G_j$ is the abstract isomorphism $P_i \mapsto P_j$. Ψ is a finite set of F_q-rational isogenies, closed under duals and **fixed before the instance is drawn**. The challenge is $X = xP_1$. CJ-DLP$_J$ is the problem: given $(P_j, X_j = xP_j)$ for $j \in J$, find x.

Definition 1 (AGM-Iso(Ψ)). The adversary A is any algorithm. It receives the curves, the P_j, the names of the isogenies in Ψ and X, all as coordinates. The game keeps a list L_j of the points of E_j that A has received; initially $L_1 = (P_1, X)$ and $L_j = (P_j)$. A query $O(\psi, Z, z)$ returns $\psi(Z)$ and appends it to $L_{t(\psi)}$. A is **oracle-algebraic** if every query has $Z = \Sigma_t\, z_t\, L_{s(\psi)}[t]$. A outputs $x^* \in Z/r$. Its internal computation is unconstrained: rho, index calculus, Semaev polynomials, Weil descent, Gröbner bases and its own isogeny evaluations are all allowed. **COM(Ψ)** is the same model without the algebraicity requirement.

This is the Fuchsbauer–Kiltz–Loss AGM, applied to a family of groups with one representation list per curve. Oracle queries count as outputs, as in AGM-plus-ROM analyses. With $\Psi = \emptyset$ every DLP algorithm is algebraic, so in the AGM DLP hardness is an assumption and never a conclusion.

3. Elliptic lemmas

Lemma 1. For F_q-rational $\psi: E_i \longrightarrow E_j$, $\psi(G_i) \subseteq G_j$ and $\psi = \lambda_\psi \cdot \tau_{ij}$ on G_i for a unique $\lambda_\psi \in Z/r$. If $r \nmid \deg \psi$, then $\lambda_\psi \neq 0$.

Proof. ψ commutes with π, so it maps rational points to rational points and r-torsion to r-torsion. A homomorphism between cyclic groups of order r is a scalar. If $r \nmid \deg \psi$, then $\ker \psi$ meets G_i trivially. □

Lemma 2 (scalar calculus). (a) $\lambda_{\psi_2\psi_1} = \lambda_{\psi_2}\lambda_{\psi_1}$. (b) $\lambda_{\hat\psi}\cdot\lambda_\psi = \deg \psi \bmod r$. (c) If $\alpha \in \mathrm{End}(E)$ for ordinary E, with $\alpha = (a + b\pi)/f$ and $r \nmid f$, then $\lambda_\alpha = (a + b)/f \bmod r$. (d) Normalise $P_j := \gamma_j(P_1)$ along a spanning tree with $r \nmid \deg$ on its edges. Then tree edges have $\lambda = 1$, their duals have $\lambda = \deg$, and any other ψ has $\lambda_\psi = \lambda_\alpha/\deg \gamma_j$ with $\alpha = \hat\gamma_j \psi \gamma_i \in \mathrm{End}(E_1)$.

Proof. (a) Evaluate both sides at P. (b) $\hat\psi\psi = [\deg \psi]$. (c) $f\alpha = a + b\pi$, and π is the identity on G, which consists of rational points. (d) Combine (a), (b) and (c). □

Lemma 3 (cofactor split). Let $s \cdot h \equiv 1 \bmod r$ and $e_G = sh \bmod N$. Then $[e_G]Z \in G$, $Z - [e_G]Z \in E(F_q)[h]$, and $\psi(Z) = \lambda_\psi \tau([e_G]Z) + \psi(Z - [e_G]Z)$.

Remark (hypothesis implicit in Thm 15.1). $[x]$ is a rational isogeny of degree x^2 with $\lambda = x$. The oracle family must therefore be independent of the instance. Thm 15.1 meets this because the adversary must name its isogenies.

4. Main results

Theorem 1 (oracle elimination). Let A be oracle-algebraic in AGM-Iso(Ψ), with Q queries, representation length $\le \ell$, time T and success ε. Let J be the set of curves touched by Ψ, together with E_1. Then there is an oracle-free B, given the instance-independent scalars λ_ψ, that solves CJ-DLP$_J$ with success exactly ε. B runs in time $T + O(Q\ell)$ operations in $Z/r + O(Q \log r)$ group operations, and never evaluates any ψ.

Proof. B stores (u_t, v_t) with $L_j[t] = u_t P_j + v_t X_j$. For a query $(\psi\colon E_i \longrightarrow E_j, Z, z)$ it computes $(u, v) = \Sigma\, z_t(u_t, v_t)$. By algebraicity $Z = uP_i + vX_i$. By Lemma 1 and $\tau(X_i) = x\tau(P_i) = X_j$, $\psi(Z) = \lambda_\psi(uP_j + vX_j)$. B returns this point and stores $(\lambda u, \lambda v)$. Every answer equals the true oracle's, so A's view is identical to the real game. □

Theorem 2 (converse). If Ψ contains a spanning tree of J with $r \nmid \deg$ on its edges, and the generators are normalised as in Lemma 2(d), then CJ-DLP$_J$ reduces to oracle-DLP using $|J|-1$ queries. Hence oracle-DLP on E_1 and CJ-DLP$_J$ are equivalent.

Proof. Query the tree edges outward from X. This returns $\gamma_j(X) = X_j$. □

Corollary 3 (computable transport). If the tree kernels are known, oracle-DLP on E_1 reduces to DLP on E_1 with success preserved. The additive loss is $\Sigma_{\text{tree}}\, C_{\text{eval}} + O(Q(\ell + \log r))$, where $C_{\text{eval}}(\ell) = O(\ell)$ with Vélu or $\tilde{O}(\sqrt{\ell})$ with √élu per prime step. Kernel finding is instance-independent preprocessing.

Corollary 4 (horizontal classes; heuristic path finding). If every curve of J has the same endomorphism ring, a path $E_1 \longrightarrow E_j$ is instance-independent. It is found in heuristic $\tilde{O}(q^{1/4})$ time (Galbraith 1999; Galbraith–Hess–Smart 2002), after which the per-instance loss is polynomial. Jao–Miller–Venkatesan reduces to a *random* curve of the level, not to a given E_j, so it is not what is used here. Vertical moves across a conductor gap with a large prime factor are where Theorem 1 ends at CJ-DLP.

Proposition 5 (unrestricted queries). COM(Ψ) is equivalent to oracle-free coordinate access plus transport oracles τ_{ij} on *arbitrary* points of G_i, plus constant-size instance-independent tables on $E(F_q)[h]$ (Lemma 3). With computable transport, COM-DLP reduces to DLP on E_1 with loss $Q \cdot \max_j C(\gamma_j) + O(Q \log N)$, which grows with Q × degrees. Unlike AGM-Iso, COM moves non-algebraic points such as factor-base points between curves. Whether this ever helps beyond CJ-DLP is **open**; §14.4 gives a heuristic argument that it is neutral at best.

model	reduces to	additive loss	where
AGM-Iso(Ψ)	CJ-DLP$_J$	$O(Q(\ell + \log r))$; no degree dependence	Thm 1, tight by Thm 2
AGM-Iso, tree known	DLP on E_1	$\Sigma_{\text{tree}}\, C_{\text{eval}} + O(Q(\ell + \log r))$	Cor 3
AGM-Iso, horizontal	DLP on E_1	$\tilde{O}(q^{1/4})$ once per curve (heuristic), then poly per instance	Cor 4
COM(Ψ), tree known	DLP on E_1	$Q \cdot \max C(\gamma_j) + O(Q \log N)$	Prop 5
COM(Ψ), no transport	none known	—	open
GGM + isogeny oracles	lower bound	$\Omega(\sqrt{r})$	Thm 15.1

Where the elliptic structure is used. The homomorphism property linearises the answers. Prime order and rationality of G make $\psi|G$ a scalar times a transport. The dual identity $\hat{\psi}\psi = [\deg]$ makes duals free. Because π fixes G pointwise, the eigenvalues of endomorphisms in the class are computable. The idempotent split $E(F_q) = G \oplus E[h]$ separates a tabulable part from the transport.

5. Summation polynomials in the model

Let $\varphi\colon E \longrightarrow E'$ be rational, $M = \ker\varphi \cap E(F_q)$, $\mu = |M|$. For a base F, $d_F^{(m)}(R)$ counts ordered m-tuples of F summing to R, and $w_F(Q') = \#(\varphi^{-1}(Q') \cap F)$.

Theorem 6 (pushforward). For every Q': $\Sigma_{\varphi(R)=Q'}\, d_F^{(m)}(R) = (w_F)^{*m}(Q')$. (a) If φ is injective on F: $\Sigma_{t\in M}\, d_F(R+t) = d_{\varphi(F)}(\varphi(R))$. (b) If $F + M = F$: $d_F(R) = \mu^{m-1}\cdot d_{\varphi(F)}(\varphi(R))$ for every R.

Proof. $d_F = 1_F^{*m}$, and pushforward along a homomorphism commutes with convolution. In (b), translating one summand by $t \in M$ is a bijection between decompositions of R and of $R + t$. □

Theorem 7 (saturation law = image law transported). Assume $r \nmid \deg\varphi$ and $F + M = F$ with $|F| = B$. (i) Exactly, $D_m^{G}(F) = D_m^{G'}(\varphi(F))$. (ii) Heuristically, $\varphi(F)$ sits in an image of index μ on E', so $c(\varphi(F)) \approx \mu\cdot c_{\text{plain}}$. (iii) Hence $c(F) \approx c_{\text{plain}}/\mu^{m-1}$, and the mitm cost factor is $(1/\mu)\cdot\mu^{m-1} = \mu^{m-2}$. These are Prop. 14.1(b) and the $|M|^{m-2}$ term of Prop. 15.2.

Proposition 8 (image bases need no oracle). Let $H = \psi(E'(F_q))$ with $r \nmid \deg\psi$. Then $R \in H$ iff $(1 - e_G)R \in H \cap E(F_q)[h]$, a fixed subgroup of size h/i. It equals $i\cdot E(F_q)[h]$ when that group is cyclic. Membership therefore costs one scalar multiplication and a lookup, so the image gain i, the automorphism fold $|A|$ and the saturation penalty are all coordinate-level effects that need no oracle.

Proposition 9 and Example (constants vs exponents). In AGM-Iso(Ψ), cost(oracle-DLP) $\le \min_{j\in J}$ cost(DLP on E_j) + ($|J|-1$) queries. The oracle therefore moves an exponent iff the class has members with different best exponents. **Example:** over F_{2^n} with composite n, GHS-weak and non-weak members coexist in one class (Gaudry–Hess–Smart 2002; Galbraith–Hess–Smart 2002), so "only constants" is false in general. In GHS02 the transport was computable by an isogeny walk, so there the oracle only removes the walk cost.

Conjecture 10 (prime fields). For ordinary classes over F_p, the best known exponent is a class invariant. Oracles then change the cost of known attacks by at most $\sqrt{3}$ for rho, and by the Prop. 15.2 factor for summation-polynomial methods. Evidence: Prop. 3.2 and Thm 15.1. Proving it would need lower bounds for non-generic algorithms.

6. Computational sanity check (measured)

`cargo run --release --example r4_16_7_agm_pushforward -- --pmin 1048576 --queries 5000 --seed 7`, single-threaded; the contended wall time of about 4 s is not a result. The curves are E: $y^2 = x^3 + 2x^2 + 2x$ over F_p, $p = 1048583$, with $N = 4\cdot 262321$ (`ICV1:fp-1048583:-700:1049284:128:unk:unk:r:d62ba3a6bad8`), and $E' = E/\langle(0,0)\rangle$: $y^2 = x^3 - 4x^2 - 4x$ (`ICV1:fp-1048583:-700:1049284:10976:unk:unk:r:07d957b2581c`). φ and $\hat\varphi$ are the explicit 2-isogeny formulas of Silverman, AEC III.4.5. Their slugs are not yet registered.

check	**result (measured)**
maps on curve, homomorphism, $\hat\varphi\varphi = [2]$, $\varphi\hat\varphi = [2]$ (2000 random points each)	2000/2000 each
$\lambda_{\hat\varphi} = \deg\varphi/\lambda_\varphi = 2$ on G, with $P' = \varphi(P)$	holds
AGM simulation: random oracle-algebraic queries (2495 φ, 2505 $\hat\varphi$)	**5000/5000** agree
simulator isogeny evaluations: per instance / instance-independent / naive	1 / 1 / 5000
cofactor split $\varphi(Z) = \varphi(Z_G) + T(Z_h)$, 2000 random Z	2000/2000
fiber-sum identity, 6 base/m combinations, every Q' in the support	0 mismatches
injective law (m = 2: 95894 R; m = 3: 15540 R)	holds at all
saturated law (m = 2: 24726 R; m = 3: 1408 R)	holds at all
$D_m(G)$ saturated on E vs its image on E' (exact prediction: equal)	6195 = 6195; 360 = 360

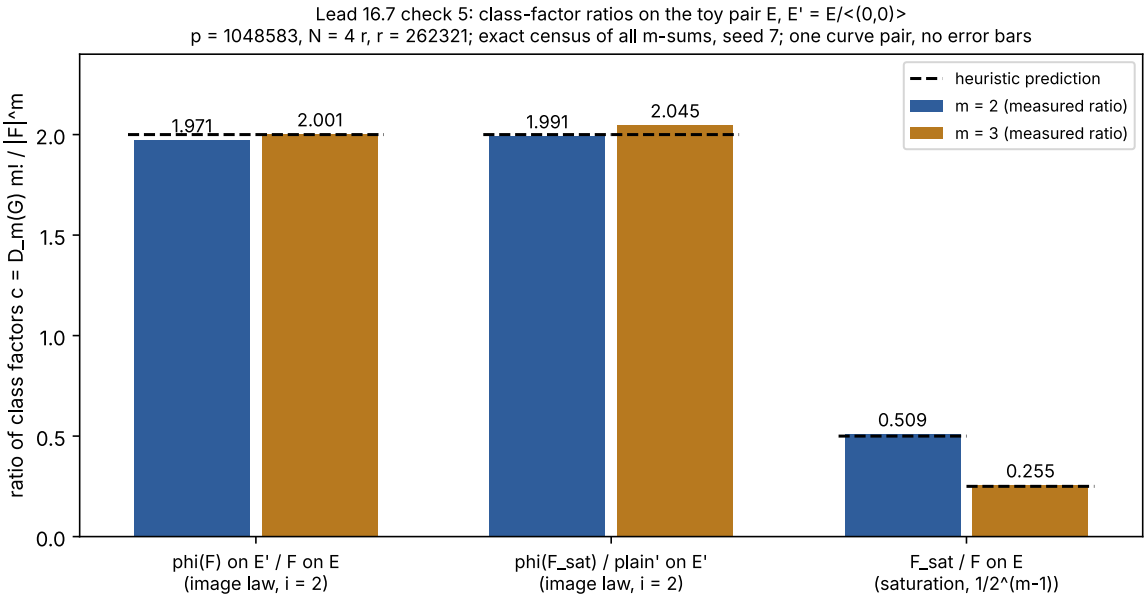


Figure 2: Ratios of class factors $c = D_m(G) \cdot m!/|F|^m$, derived from exact census counts recorded in `out_seed7.json` (check 5), against the heuristic predictions (dashed). One curve pair and one seed; no error bars are available. Base sizes: 316/158 at m = 2 and 36/32/16 at m = 3.

The plain base happened to be φ-injective at this x-range, so the "plain" and "injective" rows coincide. This is recorded as it fell, not engineered.

7. What this does not say

- It is not a hardness result. The AGM gives no lower bound for DLP, and Thm 15.1 gives $\Omega(\sqrt{r})$ only generically.
- It does not separate index calculus from rho; both are admissible adversaries.
- It does not cover non-algebraic oracle queries unless the transport is computable (Prop. 5).
- It does not settle CJ-DLP against DLP on E_1 when transport is not computable (large-prime conductor gaps).
- "Only constants" is not a theorem. It is false for GHS classes in characteristic 2 and a conjecture over prime fields.
- The class-factor ratios are heuristic and come from one toy curve pair. Theorem 6, Theorem 7(i) and Proposition 8 are exact.

8. Related work

Confident: Shoup (EUROCRYPT 1997); Nechaev (1994); Maurer (IMA C&C 2005); Fuchsbauer–Kiltz–Loss, *The algebraic group model and its applications* (CRYPTO 2018); Fuchsbauer–Plouviez–Seurin (EUROCRYPT 2020, AGM + ROM); Bauer–Fuchsbauer–Loss (CRYPTO 2020); Semaev (ePrint 2004/031); Gaudry (J. Symb. Comput. 2009); Diem (Compositio 2011); Galbraith–Gaudry, *Recent progress on the ECDLP* (DCC 2016); Gaudry–Hess–Smart (J. Cryptology 2002); Galbraith–Hess–Smart (EUROCRYPT 2002); Galbraith (LMS JCM 1999); Jao–Miller–Venkatesan (ASIACRYPT 2005); Vélu (1971); Bernstein–De Feo–Leroux–Smith (ANTS 2020); Silverman, AEC. **Recalled, details unverified:** Menezes–Teske–Weng, *Weak fields for ECC* (CT-RSA 2004); Agrikola–Hofheinz–Kastner (EUROCRYPT 2020); Zhandry, *To label, or not to label* (CRYPTO 2022); Duman–Hartmann–Kiltz–Kunzweiler–Lehmann–Riepel, *Generic models for group actions* (PKC 2023), which concerns group actions on curves, not point-evaluation oracles. **Not searched:** no web search was run. We know of no prior AGM formulation with point-evaluation isogeny oracles, but that is a statement of the author's knowledge only.